WaterScience AI Builder - Take-Home Assignment

About this assignment

Pick **one** of the four real WaterScience problems below and build a working solution in **48 hours**. You are expected to research us the way any serious candidate would, and let that shape what you build.

General Guidelines:

- **Research first:** Spend time on our website, Amazon listings and reviews, Instagram and LinkedIn. Understand our products, our customers and the problems they talk about.
- **Effort:** Plan for about 3–4 focused hours. You have 48 hours so you can think, not so you can build more.
- **Tools:** Use anything you like, including AI coding assistants (Claude, Cursor, ChatGPT, Copilot). We use them too :)
- **Data:** Create your own dummy data, based on what you learn in your research. Make it realistic enough to show your business acumen.
- **Briefs are short on purpose.** Deciding what to build is part of the test. If something is unclear, make a reasonable assumption and write it down.

How you'll be evaluated

We evaluate three things: **how well you understood WaterScience and its customers** from public sources, **whether you understand every decision in what you built** (why this design, why these calls, what it costs, where it breaks), and **whether you thought beyond the brief** about who will use it and what value it adds. A simple solution you fully understand beats an impressive one you can't explain.

What a good response looks like

	Good response	Weak response
Research	Built on what real WaterScience customers say and buy; dummy data reflects that	Generic solution that could be for any company
Starting point	Starts from who uses it, where, and on what device	Starts from which model or framework to use
Scope	Builds what the brief asks, then covers what it missed (edge cases, failure, what happens next)	Builds exactly what was written and stops
Use of AI	Uses AI only where it adds value; plain code or rules everywhere else	Adds AI everywhere to look impressive
Design	Few calls, clear data flow, knows cost and speed per run	Many chained calls nobody can justify
Scale	Knows what breaks first at 10x volume	Works for the demo only
Ownership	Can explain every line, including AI-generated code	"The AI wrote that part"
Honesty	Lists assumptions, known gaps and what to improve next	Presents it as finished and perfect
Summary	One page: what you learned about us, the user, what you built, key decisions, limits, next steps	A feature list or a copy of the README

It does not need to be perfect. It needs to be yours, and you need to know where it falls short.

The problems (pick any one)

WaterScience is a Bengaluru-based D2C brand that helps Indian homes deal with hard water. We sell products covering almost all water points at your home with specialized products such as shower filters, appliance filters, kitchen filters and whole-house filtration and water softening products on our own website, Amazon, quick-commerce apps and through retail partners. All five problems are real. Each one includes a suggested starting point for your research.

1. Plumber referral tracking

Plumbers recommend WaterScience products to customers, who receive 10% off when purchasing via them through our Shopify store. As we scale to 1,000+ plumbers, we need to attribute orders to the right plumber and track the revenue they generate. Creating and managing individual coupon codes for every plumber is not a scalable approach.

Build a system to solve this.

Where to start: Research Shopify referral and affiliate tracking, and the role plumbers play in India's bathroom products market.

Objective: Track and reward plumber-driven sales at scale.

2. Customer service overload

Our service team receives around 2000 emails and WhatsApp chats in a day: complaints, installation requests, product questions, refund requests and more. Customers wait too long for a response.

Help the service team automate common customer queries and reduce response times.

Where to start: Our product pages, and the reviews and customer questions on our Amazon listings. They show what customers actually ask and complain about. Generate synthetic customer conversations to prototype and test your solution.

3. High returns on Amazon

One of our products has a high return rate on Amazon, and its reviews are mixed. We need to know why customers are returning it.

Find out why, and tell us what we should do about it.

Where to start: The reviews on our Amazon listings. Pick any of our products with mixed reviews and treat it as the product in question.

4. Same questions, again and again

Our retail, service and operations teams keep asking the same questions about products, policies and processes. The answers exist in SOPs, product specs and policy documents, but nobody can find them quickly.

Fix this.

Where to start: use our public product pages, manuals and policies (warranty, returns, installation) as a stand-in for internal documents.

What to submit:

Reply to this email within 48 hours of receiving this assignment with:

1. **GitHub repository link** (public, or shared with us). Include the code, your dummy data, and a README that tells us how to run it.
2. **One-page summary** (PDF or link) covering:
    - what you learned about WaterScience and its customers, and how it changed your design
    - who the solution is for and what you built
    - your key technical and product decisions and why
    - assumptions, limits and what you would improve next
3. **Problem number:** Tell us which problem you chose.

A live demo link or a short screen recording is welcome but not required.

Good luck!